COMM 5559 HW 2 – Accounting Information and Stock Price

Accounting serves an important role in firm valuation. In this exercise, we will examine the relation between firm information and the stock price of public firms.

To complete this exercise, you will need data from Compustat, which can be downloaded from WRDS.

Compustat data:
Please download Compustat data (North America, Fundamentals Annual) between 2001 and 2025. The following data items are useful:

- Company ID (gvkey)
- Fiscal year (fyear)
- Total assets (at)
- income before extraordinary items (ib)
- Common equity (ceq)
- Closing stock price (prcc_f)
- Common shares outstanding (csho)
- SIC industry classification code (sic)
- R&D expenses (xrd)
- Advertising expenses (xad)
- Stock exchange code (exchg)

In addition, please apply the following rules:

- Total assets > 100 (million)
- Closing stock price > 1 (dollar)

Downloading the data in a CSV file is recommended.

Note: You might see repeated firm-years. There are several reasons for this, including for example, a change in fiscal year end. For this analysis, let's ignore this issue. We will discuss this problem in more detail in future classes.

Analysis:
We will estimate the following linear regressions:
Regression 1:

$$P = \beta_0 + \beta_1 E + \beta_2 BV + \epsilon$$

Regression 2:

$$P = \beta_0 + \beta_1 E + \beta_2 BV + \beta_3 RD + \beta_4 AD + \epsilon$$

where P is closing price (prcc_f), E is earnings per share (defined as ib scaled by csho), BV is book value per share (defined as ceq scaled by csho), RD is R&D expenses per share (defined as xrd scaled by csho), and AD is advertising expenses per share (defined as xad scaled by csho).

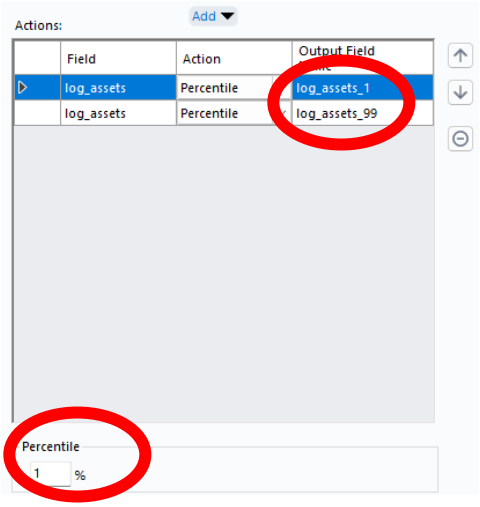
To estimate these regressions, you will need to create the variables and perform the necessary data cleaning. My recommendations are:

1. Choose the right data types.
2. Ensuring at, prcc_f, ib, csho, ceq are non-missing, because these are critical data items that shouldn't be missing for a valid data record.
3. Setting missing xrd and xad to 0, because if a firm does not have these activities or they are immaterial, Compustat often assigns missing.
4. Excluding financial service firms (sic starting with 6), because they are highly regulated and their accounting methods are different.
5. Restricting exchg to values of 11 (NYSE), 12 (NYSE American, formerly American Stock Exchange), or 14 (NASDAQ).
6. Creating the variables E, BV, RD, and AD according to their definitions.
7. Winsorize all the regression variables (e.g., P, E, BV, RD, and AD; not the raw variables such as ib and csho).

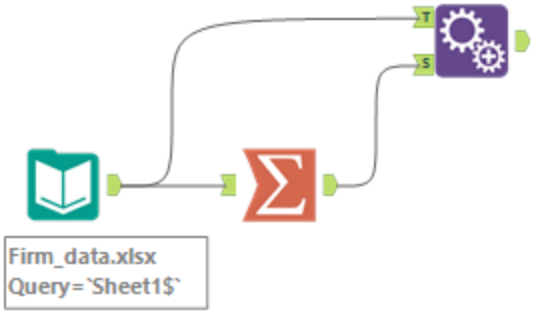
Winsorization is a statistical technique that limits extreme values or outliers in a dataset by replacing them with the top and bottom 1 percentiles (see details below).

In particular, if the value of a variable is above its top 1 percentile, then we replace it with the top 1 percentile value, and if the value is below its bottom 1 percentile, then we replace it with the bottom 1 percentile value.

To calculate the top and bottom percentiles, we can use a summarize tool. The screenshot below shows the configuration of the summarize tool. Pay attention to the bottom left corner, where we specify the percentile (1 for the bottom percentile, and 99 for the top percentile). It's also helpful to set your output field name to describe which percentile it is.



After this step, we can append the percentiles to the main data file using an "Append Fields" tool (under the Join tab). The main data file should connect to the T (target) input anchor, and the summarize tool should connect to the S (source) input anchor, like below.



The appended data will allow you to compare values with the percentiles and replace them as necessary. You can use a formula tool and IF statements or IF functions to perform the winsorization.

8. Creating an indicator for "large" firms. In particular, use a Summarize tool to calculate the median total assets (at) by year. If a firm's total assets exceed the yearly median, we set the indicator to 1; otherwise, 0.

Once the data is clean, add an Output Data tool to save it for analysis. This step is critical for automating Analysis 3.

Please create a new workflow for each analysis below.

Analysis 1:
Run regressions 1 and 2. What do you find? Are the accounting items value-relevant? What is the overall value relevance as reflected in R^2?

Analysis 2:
Run regression 2 separately for large and small firms. Is accounting information more value-relevant for large firms or for small firms? Can you guess why?

Analysis 3:
Now, we need to explore the main research question: Have accounting information lost its relevance in recent years?

Run regression 2 for all firms (i.e., do not separate large firms from small firms) for each year individually. Save the R-squared value from the yearly regression in an Excel file and create a line chart (with a trendline) to show its trend in the past 25 years. Do you see an overall decline or an increase in value relevance? Also, focus on the recession years (e.g., around 2007-2008 and around 2020-2021), do you see higher or lower value relevance? Can you explain why?

Note: Running regressions year-by-year manually is tedious. See the notes at the end for instructions on how to automate this process. We will also discuss this in class.

Deliverables:
1. Alteryx workflow file (5 in total = 1 for data cleaning + 1 for Analysis 1 + 1 for Analysis 2 + 2 for Analysis 3).

2. Word document showing a screenshot of each workflow, followed by the discussions required above (brief discussion is sufficient). In addition, attach the line charts to the Word document.
3. (Optional) Excel file.

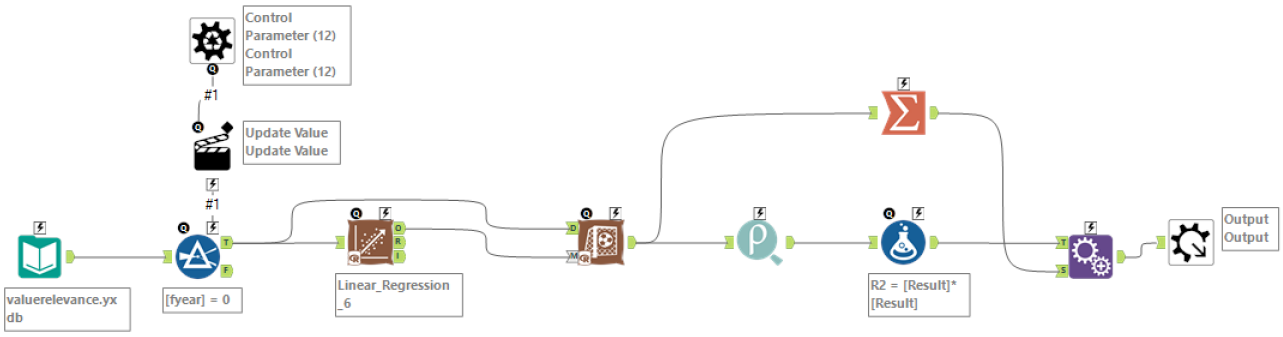
Automating Analysis 3:
We will create a batch macro to automate this process. A macro is a saved workflow or group of tools packaged into a single, reusable custom tool that you can insert into other workflows.

Thus, we can design a macro based on a simple workflow that runs the regression for a particular year (with the filter tool) and reuse it for other years.

We will use the combination of the Control Paramter and Action tools to update the year value in the filter tool.

Because the linear regression tool doesn't provide R^2 as a separate output, we will use the Score tool to compute the correlation between P and predicted P. For linear regressions, the squared correlation = R^2.

The macro workflow will look like this:



After creating the macro, we will be able to "call" it in another workflow to automate the process. In particular, we will use a Generate Rows tool to create a list of years to analyze and pass it to the macro. This workflow should look like this:

